

ABILITIES — tag review

Generated from `data/tags.json` by `engine/tag_dex.js`. Every number is a count over the **65,976** team-sheet entries in the clean corpus — not an opinion about what matters.

Your moves review is applied and committed; what changed from it is at the end.

The one thing I want you to rule on first

`profitsFromHit` **is doing two different jobs and I think it should be split.**

It covers 32 abilities and 11.2% of sheet entries, and inside it are two mechanics that imply opposite decisions:

	what happens	compounds?	what the bot should do
Holder gets stronger	Stamina +1 Def, Justified +1 Atk, Electromorphosis gains Charge, Weak Armor +2 Spe	Yes — every hit makes the next worse	stop hitting it, or kill it this turn
Attacker gets hurt	Rough Skin chip, Static / Flame Body / Poison Point status, Gooley -1 Spe, Cursed Body disable	No — a flat toll each time	keep hitting it, just not with contact

This is your Bellibolt turn exactly. Discharge into Archaludon was resisted **and** handed it a free Stamina boost — that is the first row. A Rough Skin body would have been the second row and the right play would have been different.

Rough Skin alone is 3,762 entries and Stamina 1,643, so both halves are common. My proposal is `buffsHolderOnHit` and `punishesAttacker`, with the few that do both (Weak Armor, Gooley) carrying both tags. **Say yes or no and I will do it.**

Every ability tag, ordered by how much of the format it covers

`read` means a probe string for it appears in `board.js` or the damage engine. **NOT READ** means the tag is correct and nothing consumes it yet — a wiring backlog, not an error.

`profitsFromHit` — **11.2% of entries, 32 abilities — read**

Sets: the target gains something for being hit **Why:** Stamina, Weak Armour, Berserk, Anger Shell, Justified, Rattled. Task #18 -- Will's Bellibolt case

ability	entries	share
Rough Skin	3,762	5.7%
Stamina	1,643	2.5%
Cursed Body	837	1.3%
Toxic Debris	417	0.6%
Static	278	0.4%
Flame Body	114	0.2%
Illusion	63	0.1%
Poison Point	51	0.1%
Cute Charm	48	0.1%
Electromorphosis	48	0.1%
Justified	29	0.0%
Wandering Spirit	21	0.0%
Effect Spore	20	0.0%
Mummy	16	0.0%
Goey	10	0.0%
Weak Armor	9	0.0%
...and 1 more below this cut		

damageBoost — **11.1% of entries, 44 abilities** — read

Sets: x>1 damage dealt **Why:** Adaptability, Technician, Tinted Lens, Sheer Force, Iron Fist, Strong Jaw

ability	entries	share
Blaze	2,722	4.1%
Pixilate	1,448	2.2%
Torrent	1,093	1.7%
Solar Power	432	0.7%
Overgrow	362	0.5%
Technician	344	0.5%
Tough Claws	272	0.4%
Sharpness	155	0.2%
Sheer Force	94	0.1%
Huge Power	75	0.1%
Water Bubble	73	0.1%
Supreme Overlord	60	0.1%
Iron Fist	55	0.1%
Swarm	27	0.0%
Reckless	21	0.0%
Mega Launcher	20	0.0%
...and 10 more below this cut		

onSwitchInDrop — **10.0% of entries, 3 abilities — read**

Sets: stat stages on the foe at switch-in **Why:** Intimidate. Beaten by Clear Amulet and by White Herb, neither of which is checked

ability	entries	share
Intimidate	6,604	10.0%
Supersweet Syrup	13	0.0%

priorityMod — **9.0% of entries, 22 abilities — read**

Sets: order shifts for a class of move **Why:** Prankster, Gale Wings, Triage. stallIntoEncore already depends on it

ability	entries	share
Prankster	4,692	7.1%
Gale Wings	493	0.7%
Scrappy	262	0.4%
Stance Change	149	0.2%
Mold Breaker	130	0.2%
Sheer Force	94	0.1%
Infiltrator	56	0.1%
Illuminate	27	0.0%
Keen Eye	27	0.0%
Stalwart	27	0.0%
Skill Link	4	0.0%
Long Reach	2	0.0%

weatherSetter — 7.0% of entries, 8 abilities — read

Sets: weather := x on switch-in **Why:** and megaing can COST you it, which is Will's reason to decline a mega

ability	entries	share
Drizzle	2,213	3.4%
Snow Warning	954	1.4%
Sand Stream	848	1.3%
Drought	621	0.9%

preventsStatDrop — 6.7% of entries, 15 abilities — read

Sets: stat drops simply do not apply **Why:** Clear Body (2.03%), Flower Veil for the ally. Intimidate and every -1 move do nothing, so lowersTarget is worth zero into them

ability	entries	share
Flower Veil	1,465	2.2%
Clear Body	1,331	2.0%
Hyper Cutter	378	0.6%
Inner Focus	377	0.6%
Oblivious	277	0.4%
Scrappy	262	0.4%
Mirror Armor	226	0.3%
Own Tempo	44	0.1%
Illuminate	27	0.0%
Keen Eye	27	0.0%
Big Pecks	13	0.0%
White Smoke	1	0.0%

`boostsWhenLowered` — **6.7% of entries, 2 abilities — read**

Sets: +2 to a stat when any stat is lowered **Why:** Defiant (5.46%) and Competitive. The Intimidate punisher -- dropping their Attack HANDS them an attack boost, so the lead interaction inverts

ability	entries	share
Defiant	3,610	5.5%
Competitive	815	1.2%

`contactPunish` — **6.5% of entries, 14 abilities — read**

Sets: the ATTACKER pays for touching it **Why:** Rough Skin (3,739), Static, Flame Body, Poison Point, Cute Charm, Effect Spore, Mummy, Gooley. Derived by reading the handler for `checkMoveMakesContact`

ability	entries	share
Rough Skin	3,762	5.7%
Static	278	0.4%
Flame Body	114	0.2%
Poison Point	51	0.1%
Cute Charm	48	0.1%
Wandering Spirit	21	0.0%
Effect Spore	20	0.0%
Mummy	16	0.0%
Gooley	10	0.0%

typeImmunity — 5.5% of entries, 11 abilities — read

Sets: damage of one TYPE := 0 **Why:** Levitate, Water Absorb, Flash Fire, Sap Sipper. Clicking into one wastes the turn entirely

ability	entries	share
Levitate	1,785	2.7%
Lightning Rod	1,302	2.0%
Flash Fire	357	0.5%
Dry Skin	52	0.1%
Volt Absorb	33	0.1%
Earth Eater	31	0.0%
Sap Sipper	22	0.0%
Water Absorb	15	0.0%
Motor Drive	13	0.0%

healsAllyOnSwitchIn — 5.2% of entries, 1 abilities — read

Sets: restores the partner on entry **Why:** Hospitality, 5.22% of abilities and the third most common in the format

ability	entries	share
Hospitality	3,435	5.2%

stabBoost — 4.3% of entries, 1 abilities — read

Sets: STAB becomes x2 instead of x1.5 **Why:** Adaptability, 4.34% of abilities. A flat 33% damage increase on same-type moves and nothing was reading it

ability	entries	share
Adaptability	2,855	4.3%

speedCond — 3.0% of entries, 7 abilities — read

Sets: speed x2 under a condition **Why:** Chlorophyll, Swift Swim, Sand Rush, Slush Rush, Unburden, Quick Feet. Already probed for the speed order

ability	entries	share
Chlorophyll	1,151	1.7%
Sand Rush	495	0.8%
Swift Swim	325	0.5%
Surge Surfer	8	0.0%
Slush Rush	4	0.0%
Quick Feet	2	0.0%

blocksMove — 2.9% of entries, 3 abilities — read

Sets: a whole class of move fails **Why:** already derived for allySideBlockProb -- Dazzling, Armor Tail, Good as Gold

ability	entries	share
Armor Tail	1,699	2.6%
Queenly Majesty	226	0.3%

blocksStatusMoves — 2.4% of entries, 3 abilities — read

Sets: every Status-category move fails against it **Why:** Good as Gold, 2.20%. Immune to Will-O-Wisp, Taunt, Encore, Thunder Wave -- the whole 38.5% of move slots that are status

ability	entries	share
Good as Gold	1,450	2.2%
Telepathy	116	0.2%

speedOnItemLoss — 2.2% of entries, 2 abilities — read

Sets: speed x2 once its item is gone **Why:** Unburden, 2.23%. A consumed Sash or berry doubles their speed, which flips the order mid-battle and the item tracking now makes observable

ability	entries	share
Unburden	1,465	2.2%

blocksBerries — 2.0% of entries, 3 abilities — read

Sets: their berries cannot be eaten **Why:** Unnerve, 2.03%. Turns off Citrus (10.8% of items) and every resist berry on the other side

ability	entries	share
Unnerve	1,329	2.0%

redirectsType — 2.0% of entries, 2 abilities — read

Sets: draws that type to itself **Why:** Lightning Rod and Storm Drain redirect AND boost

ability	entries	share
Lightning Rod	1,302	2.0%

weatherChipImmune — 1.9% of entries, 8 abilities — read

Sets: takes no sandstorm or snow residual damage **Why:** What onImmunity actually means for Sand Veil, Snow Cloak, Overcoat and Magic Guard -- and what typeImmunity was wrongly reporting until Will asked

ability	entries	share
Sand Rush	495	0.8%
Oblivious	277	0.4%
Snow Cloak	219	0.3%
Sand Veil	135	0.2%
Overcoat	90	0.1%
Magma Armor	44	0.1%
Sand Force	18	0.0%
Ice Body	6	0.0%

disablesAttacker — **1.3% of entries, 1 abilities — read**

Sets: the move I just used is removed from MY options **Why:** Cursed Body (833 uses). Not damage and not a stat change -- it shrinks my own option set, the same shape as locksTarget from the receiving end

ability	entries	share
Cursed Body	837	1.3%

healsOnSwitchOut — **1.1% of entries, 3 abilities — read**

Sets: restores a third of max HP by leaving **Why:** Regenerator. Makes switching a HEAL, which is the strongest argument for pivoting that the switch features cannot see

ability	entries	share
Regenerator	555	0.8%
Zero to Hero	115	0.2%
Natural Cure	38	0.1%

reducesAllyDamage — **0.9% of entries, 1 abilities — read**

Sets: my PARTNER takes x0.75 **Why:** Friend Guard. Changes every damage number aimed at the partner and nothing applies it

ability	entries	share
Friend Guard	623	0.9%

accuracyMod — **0.9% of entries, 6 abilities — read**

Sets: P(hit) scaled, often gated on a weather or a category **Why:** Sand Veil (135 uses, x1.25 evasion in sand), Snow Cloak (219, in snow), Compound Eyes, Victory Star, Hustle, Wonder Skin, No Guard. Same P(hit) the kill distribution needs

ability	entries	share
Snow Cloak	219	0.3%
Compound Eyes	210	0.3%
Sand Veil	135	0.2%
Hustle	9	0.0%
Tangled Feet	2	0.0%

damageReduce — 0.8% of entries, 9 abilities — read

Sets: x<1 damage taken **Why:** Filter, Solid Rock, Multiscale, Thick Fat, Heatproof, Fluffy. Overcalling kills without them

ability	entries	share
Multiscale	353	0.5%
Solid Rock	170	0.3%
Fluffy	3	0.0%

boostsMoveClass — 0.8% of entries, 6 abilities — read

Sets: x1.2-1.5 on moves carrying ONE FLAG **Why:** Tough Claws (contact, 272 uses), Sharpness (slicing, 155), Iron Fist (punch), Mega Launcher (pulse), Strong Jaw (bite). The join partner of moveClass -- the ability names the flag, the move carries it, and no per-ability case is needed

ability	entries	share
Tough Claws	272	0.4%
Sharpness	155	0.2%
Iron Fist	55	0.1%
Mega Launcher	20	0.0%
Strong Jaw	5	0.0%

statusImmune — 0.4% of entries, 12 abilities — read

Sets: a status cannot land **Why:** Limber, Immunity, Insomnia, Vital Spirit, Water Veil, Magma Armor. onSetStatus only -- onImmunity also means weather-chip immunity and was over-capturing

ability	entries	share
Leaf Guard	75	0.1%
Water Bubble	73	0.1%
Limber	63	0.1%
Insomnia	39	0.1%
Purifying Salt	35	0.1%
Vital Spirit	3	0.0%
Immunity	1	0.0%

formeChange — **0.4% of entries, 7 abilities — read**

Sets: the species changes mid-battle **Why:** Zero to Hero (needs a switch), Illusion, Imposter, Disguise

ability	entries	share
Zero to Hero	115	0.2%
Disguise	64	0.1%
Illusion	63	0.1%
Imposter	39	0.1%

ignoresStatStages — **0.3% of entries, 1 abilities — read**

Sets: the boost multiplier does not apply, permanently **Why:** Unaware, 172 uses. Ignores the opponent stat stages in BOTH directions, so their setup is worthless and so is yours. Same parameter Darkest Lariat sets for one move

ability	entries	share
Unaware	173	0.3%

survivesFromFull — **0.2% of entries, 1 abilities — read**

Sets: a lethal hit from full HP leaves 1 **Why:** Sturdy. Identical to Focus Sash and NOT modelled anywhere -- verified 0 mentions

ability	entries	share
Sturdy	154	0.2%

invertsBoosts — **0.2% of entries, 3 abilities — read**

Sets: stat changes flip sign **Why:** Contrary and Simple, already probed for expectedBoostSign

ability	entries	share
Contrary	134	0.2%

ignoresDefenderAbility — 0.2% of entries, 3 abilities — read

Sets: suppress every defender-side ability tag for this move **Why:** Mold Breaker, Turboblaze, Teravolt. Gates typeImmunity, damageReduce, blocksMove, preventsCrit and Sturdy in one flag

ability	entries	share
Mold Breaker	130	0.2%

preventsCrit — 0.1% of entries, 4 abilities — read

Sets: P(crit) = 0 **Why:** Shell Armor and Battle Armor. Turns Flower Trick from a guaranteed crit into an ordinary hit

ability	entries	share
Disguise	64	0.1%
Shell Armor	25	0.0%

critDamageUp — 0.0% of entries, 1 abilities — read

Sets: the CRIT MULTIPLIER itself, not its probability **Why:** Sniper (Will raised it). Three separate crit parameters exist and the taxonomy had only two: probability (Scope Lens, Flower Trick), prevention (Shell Armor) and now the multiplier. Crit damage is x1.5 and Sniper makes it x1.5 again, so x2.25 total -- it was x3 in the old gens when crits themselves were x2, which is where the folklore comes from

ability	entries	share
Sniper	24	0.0%

critRatioUp — 0.0% of entries, 2 abilities — read

Sets: P(crit) raised **Why:** Super Luck and Merciless. Same parameter as Scope Lens and Flower Trick

ability	entries	share
Merciless	3	0.0%
Super Luck	3	0.0%

terrainSetter — 0.0% of entries, 5 abilities — read

Sets: terrain := x on switch-in **Why:** same shape as weather

ability	entries	share
Psychic Surge	1	0.0%

preventsSwitch — 0.0% of entries, 3 abilities — read

Sets: the foe cannot leave **Why:** Shadow Tag, Arena Trap, Magnet Pull. Already used by the playstyle classifier

No ability with this tag appears on a single sheet in the corpus — legal, but never brought.

What your moves review changed

Six things, all cases where the engine could not see a mechanic at all.

1. Clanging Scales — you were right to be suspicious. There are *three* separate dex fields carrying self stat changes and my probes read two. `self.boosts` (Close Combat) and `secondaries[].self.boosts` (Ancient Power) were covered; `selfBoost.boosts` was not, and that is where Clanging Scales keeps its Defense drop. A 110 BP spread move with an invisible drawback. Now tagged `lowersUser`.

2. HP-gated moves. Substitute (247 uses) costs 1/4, Clangorous Soul (190) costs 1/3, Shed Tail (41) costs 1/2 — and all three *fail outright* below that. Neither the cost nor the failure was modelled. The failure also inherits the simultaneity problem: you can have the HP when you choose and not when you act, because their attack resolves first.

3. Per-hit accuracy. Triple Axel rolls each hit separately, so it lands all three **73%** of the time, not 90%. Population Bomb lands all ten **35%** of the time. Dual Wingbeat and Bullet Seed roll once for the whole move. Applying accuracy once overstated the multi-accuracy ones.

4. Moves that read the target's item. Knock Off is $\times 1.5$ into an item (1,663 uses) and Poltergeist fails without one (183). On your mega-stone question — **stones do not dodge Poltergeist**. A stone is an item and the move reads `target.item`. What stones resist is *removal*: Knock Off and Trick fail to take them, and only from the matching species. Poltergeist is near-unconditional here anyway — **106 of 65,976** sheet entries hold no item.

5. Fixed-damage moves, and this is the worst one. Super Fang, Final Gambit, Endeavor, the OHKO moves and Night Shade all have base power **0** and compute damage in a callback. The engine did not misprice them — it read them as dealing **nothing at all**. 764 uses across the corpus. Super Fang halves your HP and the bot thought it was harmless.

6. Your placeholder rule is in. Anything untagged now carries an explicit `untagged` marker instead of an empty list, and is only *flagged* if it appears on more than 0.5% of sheets. Watchog never crosses that line; a genuinely missed common move would. Right now **every move, ability and item above the floor carries at least one tag**.

Icicle Crash flinch is 30% — you asked and I owed you the number directly.

Where this actually stands

7 of 139 tags are read by nothing yet. The taxonomy is ahead of the consumer, deliberately — you are reviewing it before I wire it. But that gap is the risk: this repository has a history of models that were fitted, saved, quoted and never once used in a live decision. A tag that nothing reads is that same failure in a new shape.

So when you sign this off, the next work is not more tags. It is `tags.json` feeding the damage distribution in MAGNEMITE, and `tests/test-wiring.js` proving each one ran in a real game.